MODULE 07 · DOMAIN 2 - DATA PREPARATION

Getting text out of documents

The right Python package for each format, the distractors the exam uses, and Databricks' own parser.

SECTION 2 of 6 EXAM WEIGHT 14% MODULE 07 of 40

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Match the tool to the envelope. The format decides the package. The phrase least code decides between a specialist and an all-rounder.

Format to package

PACKAGE	USE IT FOR	WHAT IT DOES
`beautifulsoup`	HTML pages	Parses HTML and pulls out the text
`pypdf, PyMuPDF`	Digital PDFs	Reads the text layer, page by page
`pdfplumber`	Tables inside PDFs	Keeps table rows and columns
`pytesseract`	Scans and photos	OCR, using the Tesseract engine
`python-docx, python-pptx`	Word and PowerPoint	Reads each Office format
`unstructured`	Mixed formats, least code	One call partitions many formats

ON THE EXAM

Learn the three the exam uses most: beautifulsoup for HTML, pytesseract for images, unstructured for mixed content with least code.

Real packages, wrong job

- **numpy, pandas:** arrays and tables of numbers. They read no documents.
- **flask:** builds web servers. It does not read web pages.
- **requests:** fetches a page. It does not parse it.
- **scrapy:** a crawling framework for whole sites. Heavy for parsing files you already have.
- **PyPDF2** is the old name. Its work continues as **pypdf**.

> ON THE EXAM
>
> The wrong options are real, popular packages. Ask what each one's job is, and the mismatch is obvious.

Digital PDF or scanned PDF?

DIGITAL

It has a text layer

- Exported from Word or a web page
- You can select text with a mouse
- pypdf or PyMuPDF read it directly
- Fast and exact

SCANNED

It is only a picture of text

- Scanned paper or a photographed page
- Text cannot be selected
- Text extractors return empty strings
- Needs OCR, such as pytesseract

THE TIEBREAKER

A PDF is a container, not a format of text. If extraction returns nothing, the file is a picture, and you need OCR.

A specialist, or an all-rounder?

SPECIALIST LIBRARY WHEN

One known format

- Every file is the same type
- You need fine control
- Performance matters per page
- You accept more code

ALL-ROUNDER WHEN

Many formats, least code

- PDFs, Office files, HTML together
- Text mixed with images and tables
- One call per file
- Framework loaders often wrap it

THE TIEBREAKER

If the options include unstructured and the scenario mixes formats or asks for the least code, unstructured is the usual answer.

ai_parse_document: the details worth knowing

ASPECT	DETAIL
Input	Binary file content, read with READ_FILES and the binaryFile format
Formats	PDF, DOC, DOCX, PPT, PPTX, and images such as JPG, PNG, TIFF
Output	Pages and elements. Tables as HTML, with bounding boxes and confidence
Figures	Optional AI-generated descriptions, and page images saved to a volume
Limit	Over 500 pages fails unless you set pageRange

ON THE EXAM

One SQL function covers what OCR, layout detection and table extraction libraries do separately, and runs in parallel over a whole volume.

A parsing pipeline in SQL

01
Read
READ_FILES from a volume, as binaryFile.

02
Parse
ai_parse_document: text, tables, figures.

03
Prepare
Chunk, or ai_prep_search in Beta.

04
Store
A Delta table in Unity Catalog.

05
Index
An AI Search index on that table.

ON THE EXAM

Parse once into a table, then chunk, extract and classify from the parsed output. Do not re-parse for every downstream job.

A Python package, or ai_parse_document?

A PYTHON PACKAGE WHEN

The question names Python

- It asks which Python package
- Options are library names
- A single format is described
- Least lines of code is the goal

AI_PARSE_DOCUMENT WHEN

The question is Databricks at scale

- Thousands of files in a volume
- Layouts, tables and figures matter
- SQL or pipeline teams
- Results must land in governed tables

THE TIEBREAKER

Answer the question asked. If every option is a Python library, pick the library. If a Databricks option appears and scale or layout matters, prefer it.

Extraction should preserve more than words

- **Structure:** headings and sections, which module 08 uses to chunk well.
- **Tables** as tables, in HTML or markdown, not flattened into run-on text.
- **Metadata:** file name, page number and section, for citations and filters.
- **Figures:** a text description, so diagrams become searchable.
- Drop boilerplate here, at extraction, as module 06 recommended.

> ON THE EXAM
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> Page numbers captured now become citations later. Metadata lost at extraction cannot be recovered at query time.

> A software company has saved its entire help centre as HTML files. The engineer wants the article text for a rag application.
>
> The requirement is to write as few lines of Python as possible.

Which Python package should the engineer use?

A pytesseract

B beautifulsoup

C numpy

D flask

◆ PAUSE AND DECIDE

Which Python package should the engineer use?

A pytesseract
OCR for images. HTML already contains text.

B beautifulsoup
Parses HTML and returns its text in a few lines.

C numpy
Numerical arrays. Reads no documents.

D flask
Serves web pages. Does not read them.

THE SIGNAL

Saved as HTML files. The format names the package: an HTML parser.

> A research team's source folder contains PDFs with embedded images and tables, plus Word documents and PowerPoint decks.
>
> The engineer wants one approach that extracts content from all of them with the least code.

Which Python package should the engineer use?

A pypdf

B python-docx

C unstructured

D scrapy

◆ PAUSE AND DECIDE

Which Python package should the engineer use?

A **pypdf**
PDFs only, and text only. Misses Word, PowerPoint and images.

B **python-docx**
Word only. Misses the other formats.

C **unstructured**
One call per file across PDF, Office, images and tables.

D **scrapy**
Crawls websites. These are local files.

THE SIGNAL

Several formats, images and tables, and least code. That combination is the all-rounder's job.

> An engineer extracts text from 10,000 PDF contracts with pypdf. Most work well, but about 800 return empty strings, although they open and look normal.
>
> Those 800 came from a records office that digitised paper archives.

What should the engineer do for the 800 files?

A Run OCR on them, for example with pytesseract

B Switch to beautifulsoup for those files

C Increase the chunk size for those files

D Switch to pdfplumber to read their tables

◆ PAUSE AND DECIDE

What should the engineer do for the 800 files?

A OCR with pytesseract

Scanned pages have no text layer. OCR reads the picture.

B beautifulsoup

Parses HTML. These are PDFs of images.

C Bigger chunks

Chunking empty strings still gives nothing.

D pdfplumber

Better at tables, but still needs a text layer.

THE SIGNAL

Empty strings, files that look normal, and digitised paper archives. That is a picture of text, so it needs OCR.

> An engineering firm has 200,000 PDF manuals in a Unity Catalog volume, full of complex tables and wiring diagrams. Its data team works mainly in SQL.
>
> It needs tables preserved and diagrams made searchable, with results in governed Delta tables for an AI Search index.

Which approach fits best?

A Loop over all the files with pypdf in a single Python notebook

B Run pytesseract over every page of every manual, then store text

C Parse each manual with beautifulsoup and write the text to Delta

D ai_parse_document on READ_FILES binary input, with figure descriptions

◆ PAUSE AND DECIDE

Which approach fits best?

A pypdf loop in one notebook

Serial, loses table structure, and ignores the diagrams.

B pytesseract on every page

OCR text only. No tables as tables, no figure meaning.

C beautifulsoup

An HTML parser. These are PDFs.

D ai_parse_document

SQL, parallel, tables as HTML, figure descriptions, into Delta.

THE SIGNAL

A volume, 200,000 files, tables and diagrams, a SQL team, governed tables. Every signal points to the Databricks parser.

What to carry forward

- **beautifulsoup** for HTML. **pytesseract** for images and scans. **unstructured** for mixed formats with least code.
- pypdf or PyMuPDF for digital PDFs. pdfplumber when PDF tables matter.
- Empty strings from a PDF mean it is a scan. Use OCR.
- At scale on Databricks: **ai_parse_document**, tables as HTML, figures described.
- numpy, flask, requests and scrapy are real packages doing a different job.

> ON THE EXAM
>
> Next: module 08 splits the extracted text into chunks, with the arithmetic that exam questions rely on.